



GOSFORD HIGH SCHOOL

2011 TRIAL HSC EXAMINATION

EXTENSION 1 MATHEMATICS

General Instructions:

- Reading time: 5 minutes.
- Working time: 2 hours
- Write using black or blue pen.
- Board-approved calculators may be used.
- Each question should be started on a separate writing booklet.
- All necessary working should be shown in every question.

Total marks: - 84

Attempt all Questions 1- 7.

Question 1

Start a SEPARATE BOOKLET

Marks

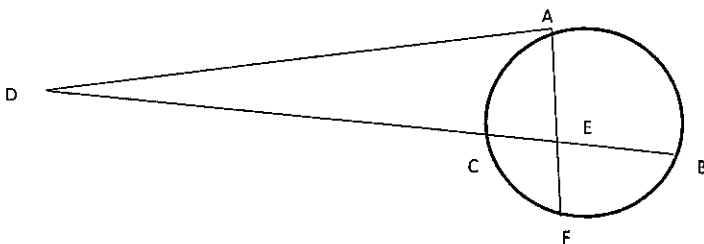
- a) Evaluate $\lim_{x \rightarrow 0} \frac{5\sin 2x}{4x}$ 1
- b) Solve $\frac{x^2-3}{2x} > 0$ 3
- c) Evaluate $\int_0^1 \frac{dx}{\sqrt{4-x^2}}$ 2
- d) Find $\int x\sqrt{x^2+1} dx$, using the substitution $u = x^2 + 1$ 3
- e) Differentiate $\log_e \left(\frac{e^x+1}{e^x-1} \right)$ with respect to x 3

Question 2

Start a SEPARATE BOOKLET

Marks

- a) In the diagram, DA is a tangent to the circle and DCB is a straight line, cutting the circle in C and B. The point E is taken on CB so that DA = DE and AE produced, meets the circle at F.



- (i) Copy the diagram into your answer booklet
- (ii) Prove that AE bisects $\angle BAC$ 3
 (hint: let $\angle DAC = \alpha$ & $\angle CAE = \beta$)
- b) (i) Show that $\frac{d}{dx} \left(\frac{v^2}{2} \right) = \frac{dv}{dt}$ 2
- (ii) When x metres from the origin, the velocity, $v \text{ ms}^{-1}$, of a particle which moves along a straight line is given by

$$v^2 = 6(16 - x^4).$$

Find its acceleration when it is at $x = \frac{1}{2}$

2

Question 2 continued

- c) P, Q and R are the points $(-5, 12)$, $(4, 9)$ and $(0, 2)$ respectively. X divides the interval PQ externally in the ratio 5:2
Prove that *angle* PRX is a right angle. 2

- d) Evaluate $\int_0^{\pi} \cos^2 2x \, dx$ 3

Question 3

Start a SEPARATE BOOKLET

Marks

- a) The area under the curve $y = \sin x + \cos x$, above the x axis and between $x = 0$ and $x = \frac{\pi}{2}$, is rotated about the x -axis. Find the volume of the solid of revolution formed. 4

- b) A disintegrating comet called Z, which is always spherical in shape, is decreasing in volume at a constant rate of $8\text{m}^3/\text{min}$. Find the rate at which the surface area is changing when the radius is 4m. 4

- c) α, β, γ are roots of the polynomial equation $2x^3 - 4x^2 + 5x - 3 = 0$. Find:

(i) $\alpha + \beta + \gamma$ 1

(ii) $\alpha\beta\gamma$ 1

(iii) $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$ 2

Question 4

Start a SEPARATE BOOKLET

Marks

- a) The line $y = mx$ makes an angle of 45° with the line $y = 2x - 3$.
Find the two possible values of m . 3

- b) The polynomial $P(x) = (x - a)^3 + b$ has a value zero at $x = 1$, and,
when divided by x , the remainder is -7.
Find all possible values of a and b . 5

- c) Use the method of mathematical induction to prove that, for all positive integral values of n for $n \geq 1$,

$$1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + n \times n! = (n + 1)! - 1$$
 4

Question 5**Start a SEPARATE BOOKLET****Marks**

- a) Sketch the graph of the function $y = 3\sin^{-1}\left(\frac{x}{2}\right)$, stating clearly the domain and range.

3

- b) (i) Given that $x^2 + 4x + 5 \equiv (x + a)^2 + b^2$ find the values for a and b. **2**

- (ii) Using the result from part (i) above, find; $\int \frac{dx}{x^2+4x+5}$ **2**

- c) A particle moves in a straight line and at time t seconds, its distance is x cms from a fixed origin point, O. The equation of the line is given by :

$$x = 1 + \frac{1}{2}\cos 2t$$

- (i) Show the motion of the particle is in Simple Harmonic Motion **1**
- (ii) State the period of motion for the particle. **1**
- (iii) Sketch the graph of x as a function of t in the domain $0 \leq t \leq 2\pi$ **1**
- (iv) Find the displacement of the particle when it is at rest and thus determine the length of its path. **2**

Question 6**Start a SEPARATE BOOKLET****Marks**

- a) (i) Assuming that $\cos x \neq 0$, make $\tan x$ the subject of $\sin(x + \theta) = a \cos x$ **3**

- (ii) Find the exact value of $\tan x$ when $\sin\left(x + \frac{\pi}{3}\right) = 2 \cos x$, and the values of x , for $0 \leq x \leq 2\pi$, correct to 4 decimal places. **2**

- b) (i) Derive the equation of the tangent to the parabola $x^2 = 4ay$ at the point P($2ap$, ap^2). **2**

- (ii) The tangent meets the line $x = a$ at Q. Find the co-ordinates of Q. **2**

- (ii) M is the mid-point of PQ. Prove that, as P moves on the parabola, M moves on a straight line. **3**

Question 7

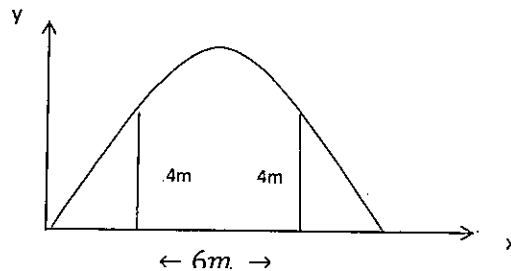
Start a SEPARATE BOOKLET

Marks

READ IT AS $f(x) = \sin\left(\frac{x}{10}\right) - e^{-x}$

- a) Find, correct to 2 decimal places, the root of $e^{-x} = \sin\left(\frac{x}{10}\right)$, using one application of Newton's Method, and taking $x = 2$ as the first approximation. 3
- b) A ball is projected from a point O with speed V m / sec and at an angle α to the horizontal. Air resistance is ignored and g m / sec² is the acceleration due to gravity.
- (i) Derive the expressions for the horizontal component $x(t)$ and the vertical component $y(t)$ of the ball's displacement after t seconds (neglect air resistance). 2
- (ii) If R is the range on the horizontal plane of this projectile, show that the cartesian equation of the path can be given by: $y = x \left(1 - \frac{x}{R}\right) \tan \alpha$ 4

(iii)



If $\alpha = 45^\circ$ and the ball just clears two vertical posts, which are both 4 metres above the level of projection and 6 metres apart, calculate the range, R . 3

END OF EXAM ☺

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

ANSWERS

2011 Ext one Trial

Q1 a) $\lim_{x \rightarrow 0} \frac{5 \sin 2x}{4x} = 5 \times \frac{2}{4}$
 $= 2\frac{1}{2}$ ①

b) $\frac{x^2-3}{2x} > 0$

Consider

$x > 0$ $x < 0$ ①

$x^2-3 > 0$ $x^2-3 < 0$

$\therefore x < -\sqrt{3} \text{ or } x > \sqrt{3}$ $-\sqrt{3} < x < \sqrt{3}$ ①

Since $x > 0$ Since $x < 0$

$x > \sqrt{3}$ $-\sqrt{3} < x < 0$ ①

c) $\int_0^1 \frac{dx}{\sqrt{4-x^2}} = \left[\sin^{-1}\left(\frac{x}{2}\right) \right]_0^1$ ①
 $= \sin^{-1}\frac{1}{2} - \sin^{-1}0$
 $= \frac{\pi}{6} - 0$
 $= \frac{\pi}{6}$ ①

d) let $u = x^2+1$
 $du = 2x$ ①

$\therefore \int x \sqrt{x^2+1} dx = \frac{1}{2} \int \sqrt{u} du$ ①
 $= \frac{1}{2} \times \frac{2}{3} u^{\frac{3}{2}} + C$
 $= \frac{1}{3} u^{\frac{3}{2}} + C$
 $= \frac{1}{3} (x^2+1)^{\frac{3}{2}} + C$ ①

e) $\nearrow$

e) $y = \log_e \left(\frac{e^x+1}{e^x-1} \right)$

$y = \log_e(e^x+1) - \log_e(e^x-1)$

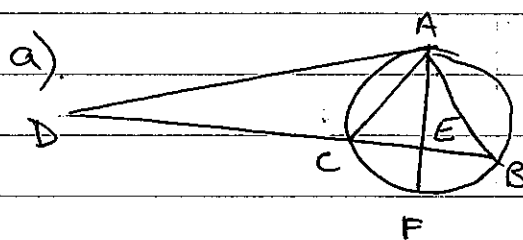
$\frac{dy}{dx} = \frac{e^x}{e^x+1} - \frac{e^x}{e^x-1}$ ①

$= \frac{e^x(e^x-1) - e^x(e^x+1)}{(e^x+1)(e^x-1)}$ ①

$= \frac{e^{2x} - e^x - e^{2x} - e^x}{e^{2x} - 1}$

$= \frac{-2e^x}{e^{2x} - 1}$ ①

Question 2



ii) let $\angle ABE = \alpha$,
 $\angle BAE = \beta$

$\therefore \angle AEC = (\alpha + \beta)$ (ext $\angle$ at ΔABE) ①

$\therefore \angle EAD = (\alpha + \beta)$ (isos ΔAED) ①

But $\angle DAC = \angle ABE$
 $= \alpha$ (angle in alt. seg) ①

$\therefore \angle EAC = \beta$

$\therefore \angle BAE = \angle EAC$
 $= \beta$

$\therefore AE$ bisects $\angle BAC$

Q 2 b

$$(i) \frac{d}{dx} \left(\frac{v^2}{2} \right) = \frac{d \left(\frac{v^2}{2} \right)}{dt} \times \frac{dv}{dx} \quad (1)$$

$$= v \times \frac{dv}{dx}$$

$$= \frac{dx}{dt} \times \frac{dv}{dx}$$

(1)

$$= \frac{dv}{dt}$$

$$\therefore \frac{d}{dx} \left(\frac{v^2}{2} \right) = \frac{dv}{dt}$$

$$(ii) v^2 = 6(16 - x^4)$$

$$a = \frac{dv}{dt}$$

$$= \frac{d \left(\frac{v^2}{2} \right)}{dx}$$

$$= \frac{d}{dx} (3(16 - x^4)) \quad (1)$$

$$= -12x^3$$

Acceleration at $x = \frac{1}{2}$

$$a = -12 \left(\frac{1}{2} \right)^3$$

$$= -1 \frac{1}{2} \text{ ms}^{-2} \quad (1)$$

C, Co-ordinate of x $\left(\frac{-5 \times 2 + 4 \times 5}{5 - 2} \right)$

$$= \left(10, 7 \right) \quad (1)$$

Gradient of PR = $\frac{2-12}{0+5}$ grad of RX

$$= \frac{7-2}{10-0}$$

$$= -2$$

$$\therefore m_{PR} \times m_{RX} = -2 \times \frac{1}{2} = -1$$

$$= -1 \quad (1)$$

$\therefore \angle PRX$ is a right angle

$$\frac{d}{dx} \int_0^\pi \cos^2 2x \, dx \quad \cos^2 x = \cos^2 x + \sin^2 x \quad (1)$$

$$= 2\cos^2 x - 1$$

$$= \frac{1}{2} \int_0^\pi (\cos 4x + 1) \, dx \quad \because \cos^2 x = \frac{\cos 2x + 1}{2}$$

$$\text{hence } \cos^2 2x = \frac{\cos 4x + 1}{2}$$

$$= \frac{1}{2} \left[\frac{\sin 4x}{4} + x \right]_0^\pi \quad (1)$$

$$= \left[\frac{\sin 4x}{8} + \frac{x}{2} \right]_0^\pi$$

$$= \frac{\sin 4\pi}{8} + \frac{\pi}{2} - \frac{\sin 4 \cdot 0}{8} - 0$$

$$= 0 + \frac{\pi}{2} - 0$$

$$= \frac{\pi}{2}$$

(1)

Question 3

$$a) V = \int_0^{\frac{\pi}{2}} \pi y^2 \, dx$$

$$= \pi \int_0^{\frac{\pi}{2}} (\sin x + \cos x)^2 \, dx \quad (1)$$

$$= \pi \int_0^{\frac{\pi}{2}} (\sin^2 x + 2\sin x \cos x + \cos^2 x) \, dx$$

$$= \pi \int_0^{\frac{\pi}{2}} (1 + 2\sin x \cos x) \, dx$$

$$= \pi \int_0^{\frac{\pi}{2}} (1 + \sin 2x) \, dx$$

$$= \pi \left[x - \frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{2}} \quad (1)$$

$$= \pi \left[\frac{\pi}{2} - \frac{1}{2} \cos \pi - 0 - \frac{1}{2} \cos 0 \right]$$

$$= \pi \left(\frac{\pi}{2} + \frac{1}{2} + \frac{1}{2} \right)$$

$$= \pi \left(\frac{\pi}{2} + 1 \right) \text{ u}^3 \quad (1)$$

Q3 Contd.

$$b) V = \frac{4}{3}\pi R^3 \quad (1)$$

$$\frac{dV}{dt} = \frac{dV}{dR} \times \frac{dR}{dt}$$

$$8 = \frac{4}{3}\pi R^2 \frac{dR}{dt}$$

$$\frac{dR}{dt} = \frac{2}{\pi R^2} \quad (1)$$

$$\text{Now } SA = 4\pi R^2$$

$$\frac{ds}{dt} = \frac{ds}{dR} \times \frac{dR}{dt}$$

$$= 8\pi R \times \frac{2}{\pi R^2} \quad (1)$$

$$= 16/R$$

$$\text{when } R = 4$$

$$\frac{ds}{dt} = \frac{16}{4}$$

$$= 4 \text{ m/min} \quad (1)$$

$$c) 2x^3 - 4x^2 + 5x - 3 = 0$$

$$i) \alpha + \beta + \gamma = \frac{-b}{a}$$

$$= \frac{4}{2}$$

$$= 2 \quad (1)$$

$$ii) \alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$= \frac{5}{2} \quad (1)$$

$$iii) \frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma} = \frac{\alpha + \beta + \gamma}{\alpha\beta\gamma} \quad (1)$$

$$= \frac{2}{2/3}$$

$$= \frac{4}{3} \quad (1)$$

Question 4.

$$a) y = mx \text{ has gradient } m$$

$$y = 2x - 3 \text{ has gradient of } 2$$

$$\tan \theta = \left| \frac{m-2}{1+2m} \right|$$

$$= \tan 45^\circ$$

$$= 1$$

$$\therefore \left| \frac{m-2}{1+2m} \right| = 1 \quad (1)$$

$$\frac{m-2}{1+2m} = 1 \quad \frac{m-2}{1+2m} = -1 \quad (1)$$

$$m-2 = 1+2m \quad m-2 = -1-2m$$

$$m = -3 \quad m = \frac{1}{3} \quad (1)$$

$$b) P(x) = (x-a)^3 + b$$

$$P(1) = 0 \quad \therefore (1-a)^3 + b = 0 \quad (1)$$

$$P(0) = -7 \quad \therefore (0-a)^3 + b = -7 \quad (2)$$

$$-a^3 + b = -7$$

$$(1) - (2)$$

$$(1-a)^3 + a^3 = 7$$

$$1 - 3a + 3a^2 - a^3 + a^3 = 7 \quad (1)$$

$$3a^2 - 3a - 6 = 0$$

$$a^2 - a - 2 = 0$$

$$(a-2)(a+1) = 0$$

$$a = 2, a = -1 \quad (1)$$

$$\therefore b = 1, b = -8 \quad (1)$$

Q4 contd

$$c) S_n = 1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + n \times n! \\ = (n+1)! - 1$$

$$\text{when } n=1 \quad S_1 = 1 \times 1! \\ = (1+1)! - 1 \\ = 1 \quad (1)$$

$\therefore$ true for $n=1$

3 Assume true for $n=k$ and prove $n=k+1$

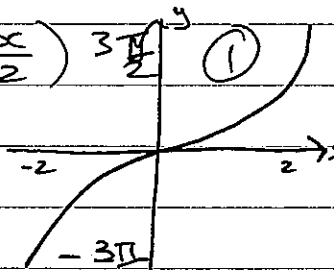
$$S_{k+1} = S_k + T_{k+1} \\ = (k+1)! - 1 + (k+1) \times (k+1)! \\ = (k+1)! + (k+1)(k+1)! - 1 \quad (1) \\ = (k+1)! \{1 + k + 1\} - 1 \\ = (k+1)! (k+2) - 1 \\ = (k+2)! - 1$$

$\therefore$ true for $n=k+1 \quad (1)$

3 since true for $n=1$, $n=k$ and $n=k+1$

$\therefore$ it is true for all positive integral values for $n \geq 1 \quad (1)$

Question 5

$$a) y = \sin^{-1} \left(\frac{x}{2} \right) \quad (1)$$


$$\text{Dom: } -1 \leq \frac{x}{2} \leq 1 \\ -2 \leq x \leq 2 \quad (1)$$

$$\text{Range: } -\frac{\pi}{2} \leq y \leq \frac{\pi}{2} \quad (1) \\ -\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$$

$$b) i) x^2 + 4x + 4 + 1 \\ = (x+2)^2 + 1$$

$$\therefore \text{If } x^2 + 4x + 5 = (x+a)^2 + b^2 \\ a=2 \quad b=\pm 1 \quad (2)$$

$$ii) \int \frac{dx}{x^2 + 4x + 5} = \int \frac{dx}{(x+2)^2 + 1} \quad (1)$$

$$= \tan^{-1}(x+2) + c \quad (1)$$

c) Replace $x-1$ by X

$$\therefore X = \frac{1}{2} \cos 2t$$

$$\dot{X} = -\sin 2t$$

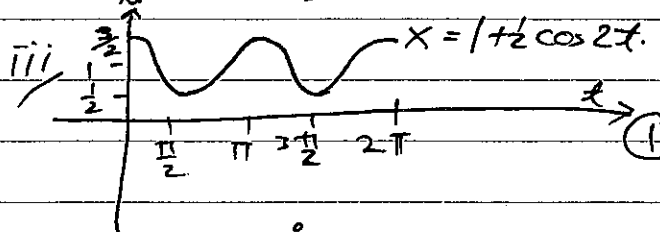
$$\ddot{X} = -2 \cos 2t$$

$$= -4 \left(\frac{1}{2} \cos 2t \right)$$

$$= -(2^2) X \quad (1)$$

$\therefore$ in SHM

$$ii) T = \frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi \text{ sec} \quad (1)$$



iv When $\dot{X} = 0$

$$-\sin 2t = 0$$

$$t = \frac{n\pi}{2} \quad n=0,1,2,3,\dots$$

$$\text{when } t = \frac{n\pi}{2} \quad X = x-1 = \frac{1}{2} \cos 2 \left(\frac{n\pi}{2} \right) \\ = \frac{1}{2} \cos n\pi \quad (1)$$

$$= \frac{1}{2} \text{ or } -\frac{1}{2}$$

$\therefore$ The displacements are $\frac{3}{2}$ or $\frac{1}{2}$ the

$$\text{length} = \frac{3}{2} - \frac{1}{2} = 1 \quad (1)$$

Question 6

$$a) \sin(x+\theta) = \sin x \cos \theta + \cos x \sin \theta$$

$$\therefore \sin x \cos \theta + \cos x \sin \theta = a \cos x \quad (1)$$

Divide by $\cos x$

$$\frac{\sin x \cos \theta + \cos x \sin \theta}{\cos x} = \frac{a \cos x}{\cos x} \quad (1)$$

$$\tan x \cos \theta + \sin \theta = a$$

$$\therefore \tan x = \frac{a - \sin \theta}{\cos \theta} \quad (1)$$

$$ii) \text{ If } \sin(x + \frac{\pi}{3}) = 2 \cos x \\ \text{then } \theta = \frac{\pi}{3} \quad a = 2 \quad (1)$$

$$\begin{aligned} \tan x &= \frac{2 - \sin \frac{\pi}{3}}{\cos \frac{\pi}{3}} \\ &= \frac{2 - \frac{\sqrt{3}}{2}}{\frac{1}{2}} \\ &= 4 - \sqrt{3} \end{aligned} \quad (1)$$

$$x = 1.1555, 4.2971 \quad (1)$$

$$b) i) x^2 = 4ay \quad \therefore y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a}$$

$$= \frac{x}{2a}$$

$$= \frac{2ap}{2a} \text{ at } (2ap, ap^2)$$

$$= p \quad (1)$$

Eqn of Tangent is

$$\begin{aligned} y - ap^2 &= p(x - 2ap) \\ y - ap^2 &= px - 2ap^2 \\ &= px - 2ap^2 \quad (1) \end{aligned}$$

$$b) ii) y - px + ap^2 = 0 \\ \text{Sub in } x = a$$

$$\therefore y - ap + ap^2 = 0 \quad (1)$$

$$\therefore Q \text{ is } (a, ap - ap^2)$$

$$iii) P(2ap, ap^2) \quad Q(a, ap - ap^2) \\ \text{Find the midpoint of M}$$

$$\begin{aligned} x &= \frac{2ap + a}{2} & y &= \frac{ap^2 + ap - ap^2}{2} \\ & & &= \frac{ap}{2} \quad (1) \end{aligned}$$

$$\text{From } y = \frac{ap}{2} \quad p = \frac{2y}{a}$$

$$\therefore x = \frac{2a(\frac{2y}{a}) + a}{2} \quad (1)$$

$$2x = 4y + a$$

$$2x - 4y - a = 0$$

ie locus of M is a straight line (1)

Question 7

a) $f(x) = \sin\left(\frac{x}{10}\right) - e^{-x}$
 $f'(x) = \frac{1}{10} \cos\left(\frac{x}{10}\right) + e^{-x}$ ①

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_2 = 2 - \frac{f(2)}{f'(2)}$$

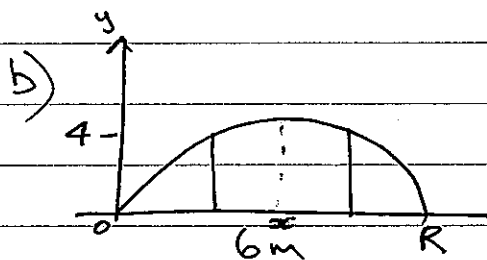
Now $f(2) = \sin\frac{1}{5} - e^{-2}$

$$f'(2) = \frac{1}{10} \cos\frac{1}{5} + e^{-2}$$

$$x_2 = \frac{2 - \sin\frac{1}{5} - e^{-2}}{\frac{1}{10} \cos\frac{1}{5} + e^{-2}} \quad \text{①}$$

$$\approx 1.073$$

①



when $t=0$ $\dot{x} = V \cos \alpha$
 $\dot{y} = V \sin \alpha$

i) $\dot{x} = 0$

$\therefore \dot{x} = V \cos \alpha$

$$x = (V \cos \alpha)t + C_1$$

when $t=0$ $x=0$

$\therefore C_1 = 0$

$\therefore x = Vt \cos \alpha$ ①

$$\ddot{y} = -g$$

$$\dot{y} = -gt + C_2$$

when $t=0$ $\dot{y} = V \sin \alpha$

$\therefore C_2 = V \sin \alpha$

$\therefore \dot{y} = -gt + V \sin \alpha$

$$y = -\frac{1}{2}gt^2 + (V \sin \alpha)t + C_3$$

①

when $t=0$ $y=0 \therefore C_3=0$

$$\therefore y = -\frac{1}{2}gt^2 + Vt \sin \alpha \quad \text{②}$$

ii/ $t = \frac{x}{V \cos \alpha}$ from ①

①

$$\therefore y = -\frac{1}{2}g \left(\frac{x^2}{V^2 \cos^2 \alpha} \right) + \frac{V \cdot x \sin \alpha}{V \cos \alpha}$$

$$y = -\frac{gx^2}{2V^2 \cos^2 \alpha} + x \tan \alpha$$

③

To find R let $y=0$

$$\therefore 0 = x \left(\tan \alpha - \frac{gx}{2V^2 \cos^2 \alpha} \right)$$

$$x = \frac{\tan \alpha \times 2V^2 \cos^2 \alpha}{g}$$

$$= \frac{2V^2 \sin \alpha \cos \alpha}{g}$$

hence

$$R = \frac{2V^2 \sin \alpha \cos \alpha}{g} \quad \text{①}$$

$$\therefore 2V^2 = \frac{Rg}{\sin \alpha \cos \alpha}$$

sub into ③

$$y = -\frac{gx^2}{\frac{Rg}{\sin \alpha \cos \alpha} \cos^2 \alpha} + x \tan \alpha \quad \text{①}$$

$$= -x^2 \tan \alpha + x \tan \alpha$$

$$y = x \left(1 - \frac{x}{R} \right) \tan \alpha \quad \text{①}$$

iii/ at

→ over

Q7 contd.

at $\alpha = 45^\circ$

$$y = x \left(1 - \frac{x}{R}\right)$$

when space halved:

$$\text{when } x = \frac{R}{2} - 3 \quad y = 4$$

$$\therefore 4 = \left(\frac{R}{2} - 3\right) \left(1 - \frac{\left(\frac{R}{2} - 3\right)}{R}\right) \quad (1)$$

$$4R = \left(\frac{R}{2} - 3\right) \left(R - \frac{R}{2} + 3\right)$$

$$4R = \frac{R^2}{4} - 9$$

$$16R = R^2 - 36$$

$$R^2 - 16R - 36 = 0 \quad (1)$$

$$(R - 18)(R + 2) = 0$$

$$R = 18 \text{ or } -2$$

as $R > 0$

$$\underline{R = 18 \text{ m}} \quad (1)$$